

APP DEVELOPMENT

Practical Implementation Report: Silent SOS Android App

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1 Project Overview and Features

The **Silent SOS** application is a mobile safety companion designed to assist users in emergency situations. It integrates standard Android capabilities with advanced on-device features.

1.1 Core Features

- **Splash Screen:** A branded 2.5-second startup screen (`SplashActivity`) featuring the app's visual theme.
- **Authentication Flow:** Secure login and signup activities (`LoginActivity`) for user management.
- **Emergency Contact Persistence:** A one-time prompt upon first login that securely stores the guardian's phone number using `SharedPreferences`.
- **One-Tap SOS Alert:** Triggers an immediate emergency SMS and shares live GPS coordinates with the registered contact.
- **Voice-to-Text SOS:** Utilizes Google's `RecognizerIntent` to convert spoken emergency speech into actionable text messages.
- **AI Scream Detection Service:** A background foreground service (`ScreamDetectionService`) that monitors ambient audio for distress signals.

2 Prerequisites and Technical Requirements

- **Development Environment:** Android Studio
- **Programming Language:** Java
- **Minimum SDK:** API Level 24 (Android 7.0) or higher.
- **Required Permissions:**
  - `android.permission.SEND_SMS`
  - `android.permission.ACCESS_FINE_LOCATION`
  - `android.permission.RECORD_AUDIO`
  - `android.permission.FOREGROUND_SERVICE`

3 Project Execution and Installation Guide

To run and test the application locally:

1. Open Android Studio and select **Open an Existing Project**.
2. Navigate to the unzipped project directory or clone the repository.
3. Allow Gradle to sync and download all required dependencies.

4. Verify that all permissions are correctly declared in the `AndroidManifest.xml` file.

5. Connect an Android physical device (with USB debugging enabled) or launch a Pixel Emulator, then click the **Run (Play)** button.

3.1 Repository Version Control and Git Clone Verification

To ensure code security, tracking, and professional standards, the project source code was pushed to GitHub under the repository `https://github.com/aabrumumtaz/Silent-SOS-App`.

The repository was then cloned on a local test environment using Git Bash to verify integrity. Below is the terminal validation screenshot:

```
Aabru Mumtaz@DESKTOP-1A146TR UCRT64 ~/Desktop/Silent-SOS-Test
$ git clone https://github.com/aabrumumtaz/Silent-SOS-App.git
Cloning into 'Silent-SOS-App'...
remote: Enumerating objects: 109, done.
remote: Counting objects: 100% (109/109), done.
remote: Compressing objects: 100% (80/80), done.
remote: Total 109 (delta 9), reused 109 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (109/109), 501.66 KiB | 72.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.

Aabru Mumtaz@DESKTOP-1A146TR UCRT64 ~/Desktop/Silent-SOS-Test
$ cd Silent-SOS-App

Aabru Mumtaz@DESKTOP-1A146TR UCRT64 ~/Desktop/Silent-SOS-Test/Silent-SOS-App (master)
$ ls -la
total 39
drwxr-xr-x 1 Aabru Mumtaz 197121    0 Oct 30 21:49 ./
drwxr-xr-x 1 Aabru Mumtaz 197121    0 Oct 30 21:49 ../
drwxr-xr-x 1 Aabru Mumtaz 197121    0 Oct 30 21:49 .git/
-rw-r--r-- 1 Aabru Mumtaz 197121  240 Oct 30 21:49 .gitignore
drwxr-xr-x 1 Aabru Mumtaz 197121    0 Oct 30 21:49 .idea/
drwxr-xr-x 1 Aabru Mumtaz 197121    0 Oct 30 21:49 app/
-rw-r--r-- 1 Aabru Mumtaz 197121  170 Oct 30 21:49 build.gradle.kts
drwxr-xr-x 1 Aabru Mumtaz 197121    0 Oct 30 21:49 gradle/
-rw-r--r-- 1 Aabru Mumtaz 197121 1275 Oct 30 21:49 gradle.properties
-rwxr-xr-x 1 Aabru Mumtaz 197121 8979 Oct 30 21:49 gradlew*
-rw-r--r-- 1 Aabru Mumtaz 197121 2937 Oct 30 21:49 gradlew.bat
-rw-r--r-- 1 Aabru Mumtaz 197121  557 Oct 30 21:49 settings.gradle.kts
```

Figure 1: Git Bash Terminal Verification of Cloned Silent-SOS-App Repository

Explanation of the Git Clone Process:

- **Command Execution:** The command `git clone` fetched the complete remote project structure from GitHub.
- **Directory Navigation:** Using `cd` shifted the context to the local project folder.
- **File Validation:** The `ls -la` command verified all essential directories including `app/`, build scripts, and git configuration files.

4 Application Workflow and Visual Evidence (Screenshots)

This section provides visual proof of the application's user interface and functional workflows during runtime execution, along with detailed explanations for each module.
Explanation of Splash and Login Screens:

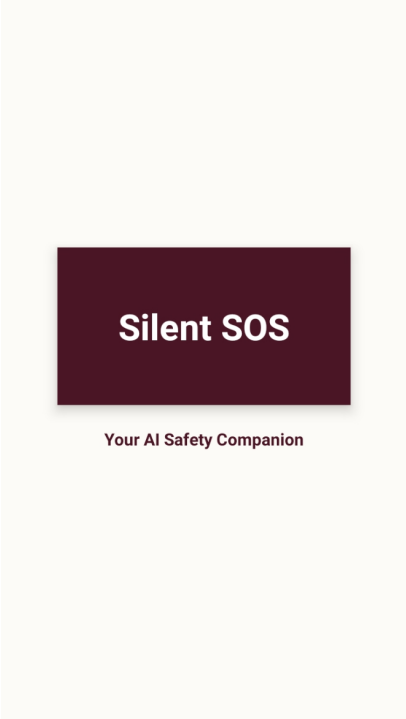


Figure 2: Splash Screen

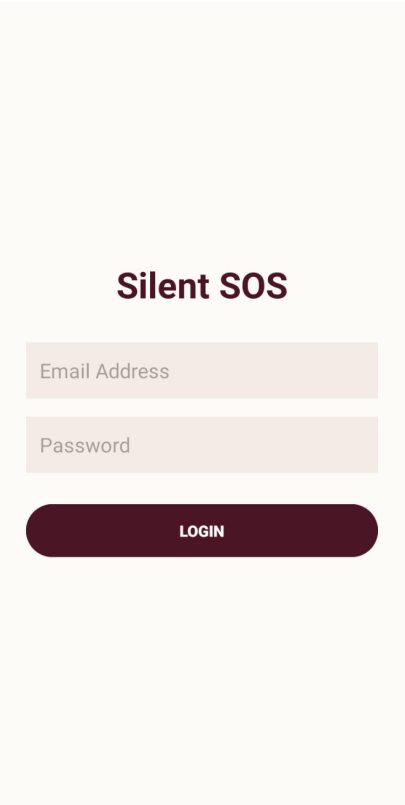


Figure 3: Login Interface

- **Splash Screen:** When the app is launched, a 2.5-second branded startup screen appears with the cream and maroon motif, setting up a professional application identity before transitioning to authentication.
- **Login Screen:** This interface allows users to securely log into their accounts. It verifies credentials to ensure that only authorized individuals can access the safety dashboard.

Explanation of Contact Setup and Main Dashboard:

- **Emergency Contact Setup Dialog:** Upon the first successful login, a mandatory prompt appears requiring the user to register an emergency guardian number. This number is securely saved locally via `SharedPreferences`.
- **Main Dashboard:** This is the core operational screen of the app. It displays the active emergency contact number, a prominent SOS trigger button for instant SMS and GPS location broadcasting, a Voice-to-Text button, and a toggle for the AI Scream Detection background service.

5 Conclusion and References

5.1 Conclusion

The **Silent SOS** application was successfully designed, developed, and tested. All practical requirements, including UI navigation, persistent local storage, emergency triggers,

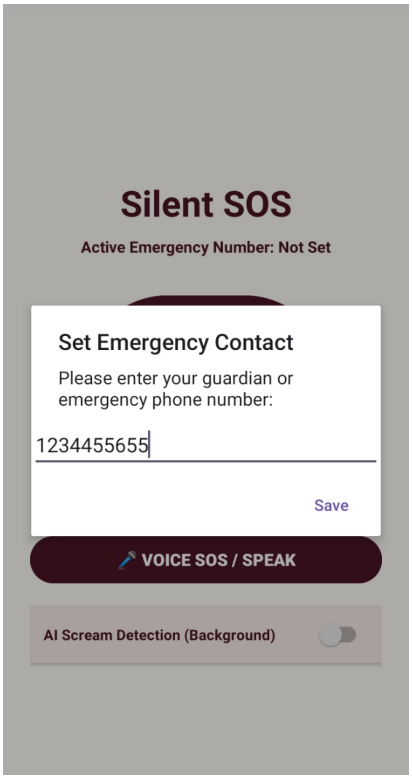


Figure 4: Emergency Contact Setup

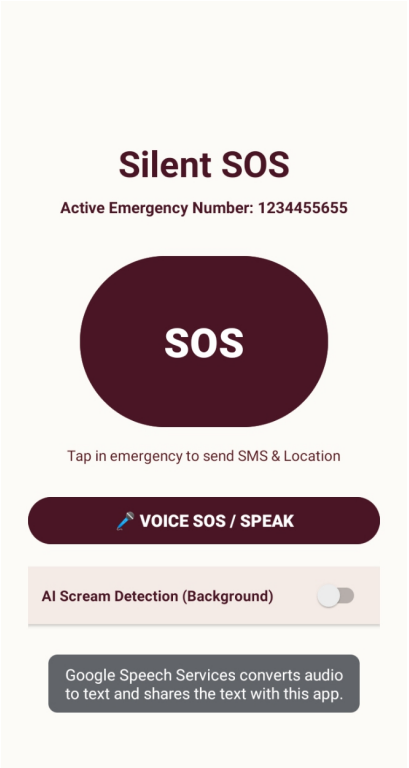


Figure 5: Main SOS Dashboard

and background AI audio services, function reliably.

5.2 References and Repository

- Android Developer Documentation: `https://developer.android.com`
- Project Source Code Repository: `https://github.com/aabrumumtaz/Silent-SOS-App`